

Exam Date & Time: 24-May-2023 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

**FOURTH SEMESTER B.TECH. (ELECTRONICS AND COMMUNICATION ENGINEERING)
DEGREE EXAMINATIONS - MAY/JUNE 2023
SUBJECT: ECE 2252/ECE_2252 - ELECTROMAGNETIC WAVES
ELECTRO MAGNETIC WAVES [ECE 2252]**

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

Missing data may be suitably assumed.

- 1A) State and prove Gauss's law in point form. (4)
- 1B) Calculate the electric flux density at point M(2, -3, 6) produced by (i) a point charge $Q = 55$ mC located at point N(-2, 3 -6); (ii) a uniform line charge $\rho_L = 20$ mC/m located on the x-axis. (3)
- 1C) A point charge Q is at the origin of spherical coordinate system. Find the flux that crosses the portion of spherical shell described by $\alpha \leq \theta \leq \beta$. What is the result if $\alpha = 0^\circ$ and $\beta = 90^\circ$? (3)
- 2A) Consider a portion of sphere given by $r = 4$ m, $0 \leq \theta \leq 0.1\pi$, $0 \leq \phi \leq 0.3\pi$.
If $\mathbf{H} = 6r \sin\phi \mathbf{a}_r + 18r \sin\theta \cos\phi \mathbf{a}_\phi$ A/m, evaluate both the sides of Stokes' theorem. (4)
- 2B) The expression for current density is given by $\mathbf{J} = 10\rho^2z \mathbf{a}_\rho - 4\rho\cos^2\phi \mathbf{a}_\phi$ mA/m².
(i) Find the current density at point P($\rho=3$, $\phi=30^\circ$, $z=2$).
(ii) Find the total current flowing outward through the circular band $\rho = 3$, $0 \leq \phi \leq 2\pi$, $2 \leq z \leq 3$. (3)
- 2C) Using Biot-Savart law determine the magnetic field intensity at any point due to an infinite element carrying current along positive z-axis. (3)
- 3A) If $\mathbf{E} = 377\cos(10^9t - 5y)\mathbf{a}_z$ V/m representing a uniform plane wave propagating in y direction in a dielectric medium, Find (i) the dielectric constant (ii) velocity of propagation (iii) intrinsic impedance (iv) wavelength. (4)
- 3B) What is skin effect? Show the graphical illustration of skin depth. (3)
- 3C) State & prove Poynting theorem. (3)
- 4A) Derive expressions for reflection and transmittance coefficient for oblique incidence of S-polarized uniform plane wave impinging on an interface between two different dielectrics. (4)
- 4B) What is Brewster's law? Derive the expression for Brewster angle. (3)

- 4C) At point P(-3, -4, 5), express that vector that extends from P to Q(2, 0, -1) in:
i) Rectangular coordinates
ii) Cylindrical coordinates (3)
iii) Spherical coordinates
- 5A) A lossless line of characteristic impedance of 50Ω is to be matched to a load whose normalized admittance $y_1 = 2 + j3.732$ by means of lossless short circuited stub. The characteristic impedance of the stub is 100Ω . Find the stub position (closest to the load) and length (using smith chart) so that a match is obtained. (4)
- 5B) Derive the expression for input impedance of a lossless quarter wave two wire transmission line. Also explain any two of its applications. (3)
- 5C) Sketch the magnitudes of voltage and current when (a) $Z_1 \ll R_0$ (b) $Z_1 \gg R_0$ (c) $Z_1 = R_0$, for a distance of 2λ from the load on a two wire transmission line at Radio frequencies. (3)

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